

Exploring Body Gestures for Small Object Selection in Dense Environment in HMD VR for Data Visualization Applications

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ABSTRACT

Recent years have seen incredible growth of Virtual Reality (VR) interfaces. In the area of data visualization and analytics, VR applications offers immense opportunities to interact, modify and explore 3D data in an interactive manner, which help in establishing new trends and patterns. In this regard, object selection is of primary importance. It is the fundamental and initial task in any immersive VR. However, the current literature is limited in investigating the effectiveness of object selection techniques in different VEs including dense and occluded dense VE, varied object sizes, proximity, and distances in the area of Immersive Analytics. In this paper, I present my ongoing PhD research to explore controller-less gestures for nail-size object selection on HMD-VR interface for dense and occluded dense VE. I describe the experiments, the findings and my future studies. I believe the outcome of these experiments in the form of guidelines or framework will enable researchers to design controller-less body gestures for object selection task.

Keywords: Virtual reality, object selection, immersive analytics, dense and occluded virtual environment, nail-size objects.

Index Terms: Human-centered computing~Human computer interaction (HCI)~HCI design and evaluation methods~User studies

1 INTRODUCTION

VR has been used in boundless application ranging from recreational use to gaming, training, therapy and education. Data visualizations in VR is one of the most promising and emerging VR application, leveraging its potential to view, explore, interact and modify 3D data in an interactive and immersive way. A few notable examples include molecular visualization, information visualization [1], scientific visualization [3], immersive workspaces [2], and embedded data representations [6].

To view 3D data in VR interfaces, object selection remains a fundamental and the primary task. Object selection is the process of indicating an object for any subsequent action like manipulation and locomotion. However, the challenges of interacting with small and occluded targets is prevalent within these VEs. In the context of immersive analytics of molecular and cellular structure that has been employed for education, communicating results, and data analysis, targets could be micro-small, occluded (far atoms by near ones), dimming distant atoms, and shadows could obscure a data point of interest from being acquired. Further, target structures needs to be pointed at, and scaled to permit the viewers (for example) to stand within a

deeply buried drug binding site inside a protein. In addition, it might be cumbersome to select and thus manipulate hidden components considering a dense environment as shown in Figure 1 and similar environments. I take this opportunity to address these issues in my thesis.

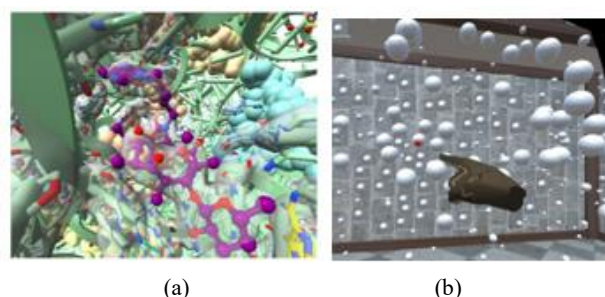


Figure 1: (a) An example of a dense environment, viewing complex scenes inside Antibiotic telithromycin [3] (b) The dense virtual environment created for the purpose of the study.

2 PROPOSED PHD RESEARCH

The research aims to investigate user-centric controller-less gestures for object selection in dense and occluded dense VEs. Moreover, it will study the influence of object proximity, object size and object distance on each of the designed selection methods. Within this view, this research presents a core research questions (RQ) and 3 sub-questions for each core RQ with an aim to pursue them during the course of this Ph.D.

RQ: Which of the new controller-less body-gestures are required for effective object selection in the dense and occluded dense VE for HMD based VR?

RQ1: How does the proximity of virtual objects influence the effectiveness of designed controller-less body-gestures?

RQ2: How does the size of the virtual objects influence the effectiveness of the designed controller-less body-gestures?

RQ3: How does the distance of the virtual objects from the users' view influence the effectiveness of the designed controller-less body-gestures?

3 EXPERIMENTS CONDUCTED AND METHODOLOGY

To investigate our RQs, we conducted a two-stage gesture elicitation study to investigate user-centric body-gestures in dense and occluded VE. In study 1, we asked 40 participants to propose the most natural and intuitive gestures while considering full body gestures for selecting a nail size object in 4 different VE conditions. The 4 VEs we designed were, (i) VE1: dense VE with the target object placed at arm's length (within 50 cm motor space), (ii) VE2: dense VE with the target object placed at scaled length (5.2 m), (iii) VE3: dense VE with the occluded target object placed at arm's length (within 50 cm

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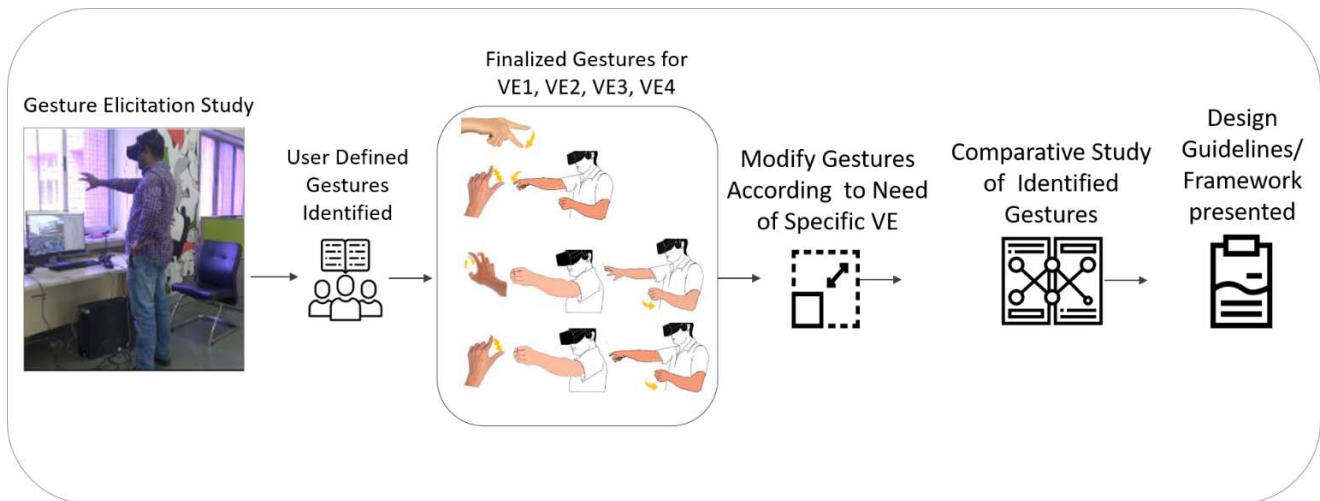


Figure 2: Design of proposed experiments

motor space) and (iv) VE4: dense VE with the occluded target placed at a scaled length (5.2 m). The task was to select the red target which was placed at one position among the grey coloured distractor targets. We created 4 dense VEs composed of 924 spheres each. Figure 1 (b) gives a representation of the VEs. Overall, 40 participants generated a total of 737 gestures for all VEs. A total of 196, 194, 171, and 186 gestures and 114 unique gestures (23, 29, 26 and 36 unique gestures) were proposed for VE 1, 2, 3 and 4 respectively. We categorized the gestures and based on a score finally extracted 3, 2, 3, and 3 gestures for all 4 VEs using an overall score (frequency of gesture + posture scores) of the performed gestures. In study 2, a new group of 40 participants evaluated the extracted gestures on ease of performance, body-part suitability, gesture appropriateness, effort and preference to finally elucidate one gesture for each VE. *Point and tap* gesture performed the best and was chosen for VE1, similarly *Pinch-out VE* and *point and tap*, *grab and rotate VE* and *point and tap*, and *pinch-out and rotate the VE* and *point and tap* were chosen for VE2, VE3, and VE4 (figure 2).

An ongoing study is a comparative study of *point and tap* gesture (gesture chosen in VE1) with existing gestures from literature *magnetic grasp* [4], *pinch* [4] and *raycasting* [5]. The task and design of VE remained same as Study 1. We improved upon the *point and tap* technique and formed a design rationale to cater to problems of accidental selection errors due to hand jitter for a task of selecting small objects within arm's reach distance in a highly dense VR environment. We will evaluate the effectiveness and efficiency of these techniques by varying the target sizes. Our study has two independent variables: *Technique (point and tap, magnetic grasp, pinch, raycasting)* and *target sizes (0.5, 2, 4 mm)* for a 4x3 design. The dependent variables are task completion time, error rate, easy to use, easy to learn and perceived naturalness of the selection method. There were 5 targets to be selected which were positioned in the comfortable zone in the VE. The order of all of the above conditions and factors are determined by Latin squares. In total, our experiment has thus $4 \times 3 \times 5 = 60$ target spheres. After the completion of the study, the results will help me to understand the gesture technique suitable for the specific VE condition.

4 EXPECTED CONTRIBUTION

As future work, I plan to conduct 3 more experiments comparing the newly designed user-centric body gestures (gesture identified

for VE2, VE3, VE4) to most commonly used body-gestures identified from the literature. These experiments will study the effectiveness – mainly error rates or accuracy of small object selection, task completion time, user preferences and ease of use. Finally, based on these results, a series of design guidelines or a framework design will be proposed as a core contribution to this thesis. The guidelines or framework will enable practitioners and/or researchers to design controller-less body gestures for object selection in a dense and occluded dense VE with varied object parameters.

5 CONCLUSION

I have a few questions that I would like to embark on this research: What context area should I consider for the gestures and how should I categorize the gesture for its applicability in the context of HCI? What other considerations should I inquire for object selection task?

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